

Business Summary Report: Predictive Insights for Collections Strategy by Aisha O Inuwa.

1. Summary of Predictive Insights

Top 3 Risk Factors:

- Missed payments: Repeated missed payments are the clearest behavioral warning sign because they directly show difficulty meeting repayment obligations.
- High credit utilization: Customers using a large share of their available credit may have less financial flexibility; the EDA also identified four values above 100%, which should be reviewed before operational use.
- High repayment pressure: A combination of high loan balance and lower income may indicate reduced capacity to keep up with repayments, especially when paired with negative payment history.

The highest-priority segment for Collections is customers showing repeated Late or Missed statuses across Month_1–Month_6, particularly when this behavior is combined with high Credit_Utilization, high Loan_Balance, or lower Income. Credit_Score and Employment_Status can provide additional context, but the model should use the full combination of variables rather than relying on one factor alone.

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Repeated negative payment behaviour is the strongest early warning signal.	Customers with multiple Late or Missed payments across Month_1–Month_6.	Missed payments, late payments, recent payment behaviour.	Prioritize early outreach and tailored repayment support before delinquency worsens.

2. Recommendation Framework

Restated Insight: Customers with repeated missed or late payments are likely to be the most useful early-warning segment for Collections, especially when combined with high credit utilization.

Proposed Recommendation: Introduce a proactive early-intervention workflow that flags customers with repeated negative payment behaviour for personalized Collections outreach.

Specific: Create a weekly watch-list for customers with two or more Late/Missed payments in the six-month payment history and prioritizes them for early contact.

Measurable: Track the share of flagged customers who return to on-time payment and compare subsequent delinquency rates with a similar non-intervention or historical baseline.

Actionable: Collections staff should contact flagged customers using approved scripts, confirm repayment difficulties, and offer appropriate payment reminders or support options.

Relevant: The recommendation targets the strongest behavioural risk signal identified in the EDA and supports Geldium's goal of reducing preventable delinquency.

Time-bound: Pilot the workflow for 90 days, review results monthly, and decide whether to scale, adjust, or stop the intervention at the end of the pilot.

Justification and Business Rationale: This approach is practical because it uses existing payment-history fields to identify customers before problems become more serious. It gives the Collections team a clear, measurable action while allowing Geldium to test whether earlier engagement improves repayment outcomes.

3. Ethical and Responsible AI Considerations

Fairness risk 1 – Biased data or inconsistent categories: Inconsistent labels such as Employed, employed, and EMP, or different patterns of missing data across groups, could create misleading model behaviour. Mitigation: standardise categories, investigate missingness, and compare model error rates across relevant customer groups before deployment.

Fairness risk 2 – Unequal false positives and false negatives: The model may incorrectly flag one group as high-risk more often, or miss genuinely high-risk customers in another group. Mitigation: monitor precision, recall, false-positive and false-negative rates across available groups and review thresholds or features if material differences appear.

Plain-language explainability: For non-technical stakeholders, the prediction should be explained as a risk estimate based on patterns such as repeated missed payments, high credit utilization, and repayment pressure—not as a final judgement about a customer.

Responsible use: The model should support human decision-making and prioritisation, not automatically impose irreversible outcomes. Geldium should document model limitations, protect customer data, monitor performance over time, and keep the process accountable and transparent.